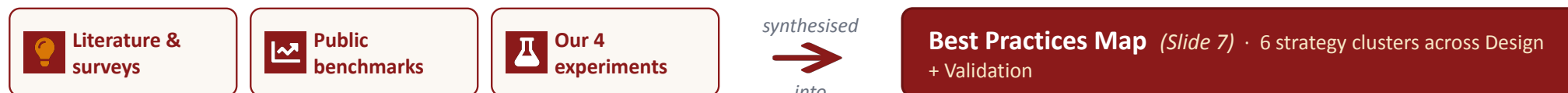


LLM-as-Judge

What Actually Works?

LLM judges power faithfulness and completeness evaluation in RAG and summarization, but reliability depends on model choice, prompt design, and aggregation. We distil best practices from prior literature, public benchmarks, and our own experiments — synthesised into a unified map (Slide 7).

OUR APPROACH



DATASETS

RAGTruth QA 5934 samples · 1724 hallucinated [Niu 2024]

HaluEval Summarization 2000 samples · 1000 hallucinated [Li 2023]

RAGBench Q&A 1500 samples · 500 incomplete

Internal data Summ 193 + RAG 295 · injected hallucinations

JUDGES & GENERATORS

Judge models Gemini 2.5 Pro · Claude Sonnet 4 · GPT-4.1

Generators (RAGTruth) gpt-3.5-turbo, gpt-4, llama-2-13b/70b-chat, mistral-7B-instruct

Prior baselines RAGTruth F1 45.6 [Niu 2024] · FaithJudge ~F1 64 [Tamber 2025] · HaluEval ~61% [Li 2023]

Experiment 1 — Do judges favor their own outputs?

BEST PRACTICES

Model selection

Structured prompting

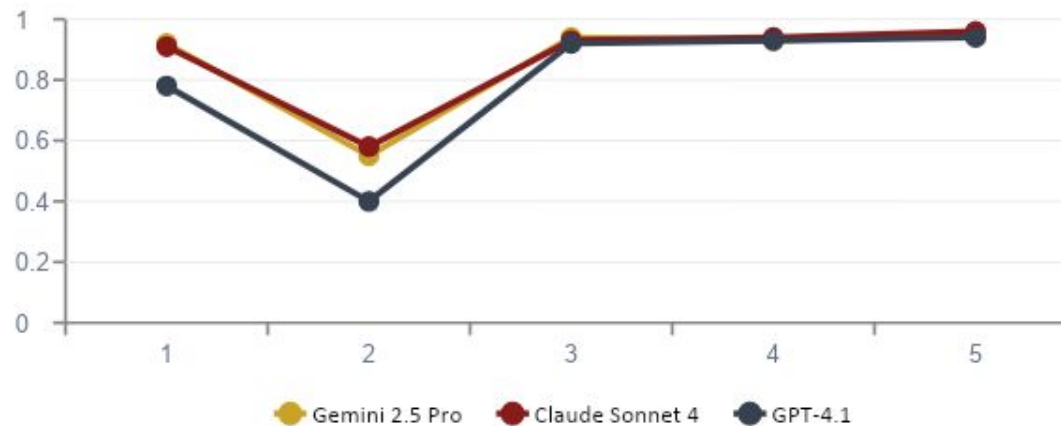
Improved task understanding

Aggregation across judges

Reporting scores

Judge evaluation

Hallucination capture rate (recall) — P4 prompt, RAG setting



! FINDING

GPT-4.1 under-flags its own hallucinations
Recall drops sharply when judging gpt-4 outputs (self-preference bias).

Cross-model judging + jury aggregation restore recall.

BEST PRACTICE · MODEL SELECTION

Frontier LLM / LLM scale

Judge $\geq$ generator capability. GPT-4/5, Claude Opus/Sonnet, Gemini Pro. G-Eval p=0.514; MT-Bench >80% agreement.

[Liu 2023; Zheng 2023]

Smaller LLMs

Fine-tuned for narrow tasks; pair with jury. Prometheus 13B Pearson=0.897; PoLL >7× cheaper than GPT-4.

[Huang 2025]

Cross-model judging

Use a judge from a different model family than the generator. Mitigates self-preference and familiarity bias.

[Wataoka 2024; Liu 2023]

Experiment 2 — Does prompt design matter?

THE PROMPT LADDER

P1 Vanilla binary

P2 + Task definition

P2* Rubric score (1–5)

P3 + Reasoning before verdict

P4* Decomposed (claims + evidence)

P4 Two-stage pipeline ★

Jury P4 × multi-model vote

P2 — Define task

A response is UNFAITHFUL if:

- info not present in source
- contradicts the source
- adds unsupported details

Return: FAITHFUL / UNFAITHFUL

P2* — Rubric score

Rate faithfulness 1–5:

- 5 = fully grounded
- 3 = partially supported
- 1 = fabricated / contradicts

Return: <score>

P3 — Add reasoning

[Task + UNFAITHFUL definition]

Briefly explain your reasoning, then provide the final verdict.

Return: {reasoning, verdict}

P4 — Worked example: claim extraction → verification → score

INPUT

Source: "The Eiffel Tower in Paris was completed in 1889."

Response: "The Eiffel Tower is in Paris, built in 1889 by Gustave Eiffel himself."

STAGE 1 • EXTRACT CLAIMS

- C1.** Eiffel Tower is in Paris
- C2.** Built in 1889
- C3.** Built by Gustave Eiffel

STAGE 2 • VERIFY + SCORE

- C1** ✓ SUPPORTED
- C2** ✓ SUPPORTED
- C3** ✗ UNSUPPORTED

supported_ratio = 2 / 3 →
UNFAITHFUL



P4 may run as a single prompt that performs both stages internally — same logic, less modularity.

Results — Each prompt layer pays off

BEST PRACTICES

Model selection

Structured prompting

Improved task understanding

Aggregation across judges

Reporting scores

Judge evaluation

FAITHFULNESS & COMPLETENESS — ACCURACY / CAPTURE RATE (%)

Approach	RAGTruth (Acc / HCR)	HaluEval (Acc / HCR)	RAGBench (Acc / ICR)
P1	66.2 / 81.9	68.9 / 69.5	—
P2	68.5 / 85.8	70.4 / 73.5	78.8 / 69.0
P2*	70.1 / 88.7	70.8 / 74.1	80.2 / 71.0
P3	70.3 / 89.0	72.7 / 76.2	—
P4*	71.2 / 91.4	74.1 / 79.1	81.0 / 75.7
P4	72.4 / 92.6	74.8 / 81.2	82.8 / 82.8
Jury (P4)	73.9 / 94.1	76.3 / 85.4	—

💡 *Monotonic climb P1 → P4 → Jury. Every prompt-ladder element from Slide 3 contributes measurable gains.*



Structured prompting

Define task + metric, chain-of-thought, JSON schemas. Drives the P2 → P3 → P4 gains.

[Gu 2024; Wei 2022]



Improved task understanding

Task decomposition (claim → evidence → verdict) and few-shot anchors. Drives the P4* / P4 gains.

[Gu 2024 RAGAS; Brown 2020]



Bias mitigation

“Ignore style, eloquence, length” counters verbosity and stylistic preference bias.

[Zheng 2023]

Experiment 3 — Jury effect: aggregation wins

BEST PRACTICES

Model selection

Structured prompting

Improved task understanding

Aggregation across judges

Reporting scores

Judge evaluation



FINDING

Multi-model Jury (P4) beats every single-judge configuration. RAGTruth: 73.9% acc / 94.1% HCR. HaluEval: 76.3 / 85.4.
Heterogeneous panels reduce correlated errors.

This motivates three practices around how we combine, report, and validate.

BEST PRACTICES · AGGREGATION · REPORTING · EVALUATION



Aggregation across judges

- Heterogeneous panels reduce correlated errors
- Majority voting, reliability-weighted
- PoLL (Haiku + Command R + GPT-3.5) $\approx$ a large judge

[Verga 2024; Dawid & Skene 1979; Whitehill 2009]



Reporting scores

- Ratio: supported claims / total claims
- Rubric 1–5 when partial quality matters
- Transparent, evidence-grounded scoring

[Es 2023 RAGAS; Gu 2024]



Judge evaluation

- Report HCR / ICR + accuracy, F1, Spearman
- Validate against human / gold labels
- Keep human annotations independent

[Zheng 2023; Liu 2023; Niu 2024; Li 2023; Tamber 2025]

Experiment 4 — Does it hold on our own data?

SETUP

Summarization 193 summaries (103 faithful, 90 with injected hallucinations). **RAG** 295 summaries (205 faithful, 90 injected). *Severity levels: subtle · moderate · aggressive. Judged with P4 prompt.*

RESULTS — ACCURACY / RECALL / F1

Severity	Summarization Acc / Rec / F1	RAG Acc / Rec / F1
<i>LLM vs Ground Truth</i>	0.8445 / 0.8351 / 0.8351	0.8920 / 0.9200 / 0.9100
Subtle	0.8039 / 0.8039 / 0.8913	0.9000 / 0.9000 / 0.9247
Moderate	0.8965 / 0.8965 / 0.9454	0.9400 / 0.9400 / 0.9691
Aggressive	0.9000 / 0.9000 / 0.9473	0.9650 / 0.9650 / 0.9822

OBSERVATIONS

Scales with severity

Overt fabrications easiest to catch (F1 0.98 RAG aggressive).

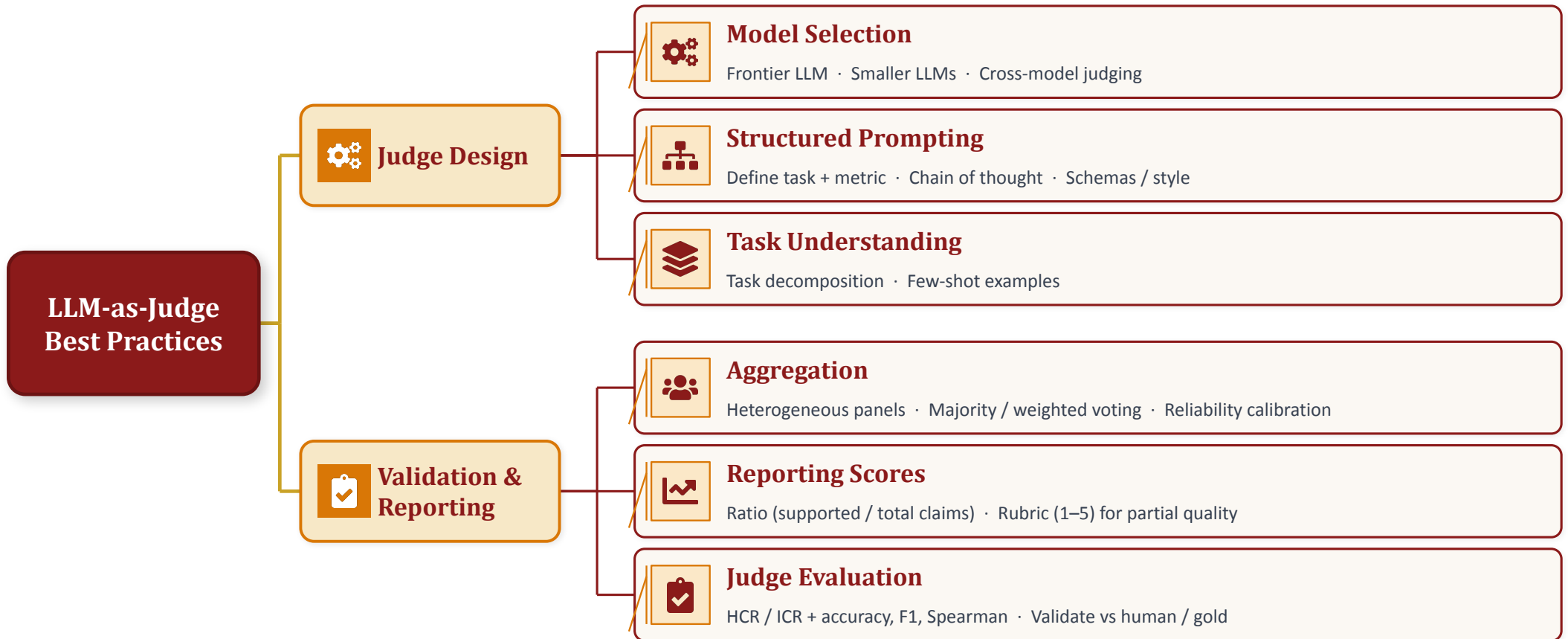
Conservative tendency

Errors skew false-positive — judge prefers safe over permissive.

Subtle remains the frontier

Plausible deviations still escape detection in a nontrivial share.

Best Practices Map



Biases addressed:



Self-preference



Verbosity



Style preference



Prompt sensitivity



False acceptance